```r
#install.packages("readxl")
library(readxl)
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(readr)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.1     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.1
v purrr     1.0.2

-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```
#install.packages("tinytex")
#tinytex::install_tinytex()
library(gt)
```

Description of dataset:

What is the data source? (1-2 sentences on where the data is coming from, dates included, etc.) How does the dataset relate to the group problem statement and question?

> The data comes from CalEnviroScreen 4.0 (2021) by California's Office of Environmental Health Hazard Assessment (OEHHA) and the California Department of Health Care Access and Information (HCAI), which provides county-level asthma emergency department visit rates from 2016–2021. These datasets would help us explore how environmental burdens across California counties relate to asthma outcomes and identify which areas could benefit from targeted interventions that could make the biggest impact.

Identify data types for 5+ data elements/columns/variables:

> 5 columns - chhs_asthma_ed: county, ED visit rate (and # ED visits?) - ces_measures: county, asthma, PM2.5, diesel.PM, poverty, pollution burden.
>
> Questions we have: - in ces_measures, what does the asthma percentile column represent? numbers seem high to be a rate of the general population.

Describe cleaning that each data source may need

> cleaning: - NA values from ces_measures: asthma, number_of_ed_visits, poverty: We will replace with 0, "Unknown", or median(df$column1, na.rm = TRUE) depending on context.
>
> We changed the census_tract column of the ces_measures to character to account for any leading zeros.

You can add options to executable code like this

The `echo: false` option disables the printing of code (only output is displayed).

> 5 columns - chhs_asthma_ed: county, ED visit rate (and # ED visits?) -
>
> ces_measures: county, asthma, PM2.5, diesel.PM, poverty, pollution burden.

In order to prepare each dataset for analysis (milestone #3), you will be expected to:

- **Cal Enviro Screen datasets**

    - **Recode common variable(s) to be combined with other datasets**
        - this means make column names match?
    - **Identify environmental measure(s) of interest and summarize into a county-level value**
        - for indicators that are broken apart by census tract, combine into one county-level number, one row per county
            - we could do median of pm2.5, diesel.PM, poverty, pollution burden
    - **Summarize CalEnviroScreen score into a county level variable**
        - ces_scores_demog dataframe: ces_4_0_score column- combine into one row per county averaging all of the values.
            - one row per county, mean of all ces_4_0_score values.
    - **NOTE: the methods for each summary variable (environmental measure and CalEnviroScreen score) should be different (i.e. if you calculate the mean for one, please use another method for the other). If you look at more than one environmental measure you can re-use summarization methods for this.**
- Asthma ED dataset - Angela
    - Recode common variable(s) to be combined with other datasets_ *done*
    - Recode values that are reading in incorrectly_ *done*
    - Subset to most recent year and county level data_ *done*
    - Select demographic strata of interest_ *demographic strata of interest included was total population, so that we can more readily compare the numbers across data sets.*
    - Pivot table to only include one row per county _ *used other functions to create one row per county.*

For all scenarios, please turn in an html document created from an Rmd or Qmd with the following components:

- Subset rows or columns, as needed
- Create new variables needed for analysis (minimum 2)
    - New variables should be created based on existing columns; for example
        - Calculating a rate
        - Combining character strings

    * Aggregation

- Clean variables needed for analysis (minimum 2)
  - Examples
    * Recode invalid values
    * Handle missing data
    * Recode categories
- Data re-structured as needed (aggregated/summarized and/or pivoted)
- Data dictionary based on clean dataset (minimum 4 data elements), including:
  - Variable name
  - Data type
  - Description
- One or more tables with descriptive statistics for 4 data elements
- Html output that is professionally prepared for presentation
  - Only the necessary information is outputted (you should suppress, for example, entire data frame outputs)
  - Use of headers and sub headers to create an organized document

Milestone 3 Rubric

Criteria	Ratings	Milestone 3 Rubric Pts
This criterion is linked to a Learning Outcome Subset rows or columns as needed	**2 ptsFull Marks** **0 ptsNo Marks**	2 pts
This criterion is linked to a Learning Outcome Create new variables needed for analysis (minimum 2)	**3 ptsFull Marks** **0 ptsNo Marks**	3 pts
This criterion is linked to a Learning Outcome Clean variables needed for analysis (minimum 2)	**3 ptsFull Marks** **0 ptsNo Marks**	3 pts
This criterion is linked to a Learning Outcome Restructure datasets, as needed	**2 ptsFull Marks** **0 ptsNo Marks**	2 pts

This criterion is linked to a Learning Outcome Data dictionary based on clean dataset (minimum 4 data elements)	**4 ptsFull Marks** **0 ptsNo Marks**	4 pts
This criterion is linked to a Learning Outcome One or more tables with descriptive statistics for 4 data elements	**4 ptsFull Marks** **0 ptsNo Marks**	4 pts
This criterion is linked to a Learning Outcome HTML document that is professionally prepared for presentation	**2 ptsFull Marks** **0 ptsNo Marks**	2 pts

Total Points: 20

Milestone #3 starts here

Data Dictionary

California County	California county that the census tract falls within

Mean CES 4.0 Score	Mean of CalEnviroScreen Score, Pollution Score multiplied by Population Characteristics Score

Median PM2.5	Annual median PM 2.5 concentrations

Median Diesel PM	Annual median Diesel PM emissions from on-road and non-road sources

Median Poverty	Median Percent of population living below two times the federal poverty level by county

Median Traffic	Median traffic density, in vehicle- kilometers per hour per road length, within 150 meters of the census tract boundary by county

Cleaning /Creating New Variables in Dataframes

Combine Dataframes and Re-Structured

Data re-structured and summarized

Create a Table for HTML Output

Data Dictionary

MILESTONE 4 STARTS HERE

Please submit an Rmd or Qmd and publish an html file on RPubs with the following:

1. Final datasets for creation of visualization
   - Join all datasets together
   - Calculate any remaining data elements needed for analysis
   - Show code used to create joined dataset, but please do not print full data frame output (showing data structure with `str()` is okay)
2. Visualizations (at least one per group member)
   - One print quality table as requested in scenario
   - One print quality plot or chart as requested in scenario
   - For groups of 3, one additional print quality table or plot of your choice (can support the requested data in the scenario, or answer a different question using the same data sources)
3. For each visual, include
   - Code used to generate visual
   - Legend (if necessary)
   - 1-2 sentence interpretation
   - NOTE:

- Please make sure the visual can stand-alone, meaning it includes enough information in title, legend, and footnote so that a person who sees only the visualization could understand what is being presented.
- Please also make sure column names, axis labels, and any other labels are meaningful and not just the name of the variable (ex: "County" rather than "county_name")

4. Html file that is professionally prepared for presentation and published to RPubs

   - Only the necessary information is in the output (e.g., suppress entire data frame outputs but showing data structure with `str()` is okay.
   - Code used to import and clean data (milestones 2 and 3) can be included in the rmd/qmd but please use the option include=FALSE in order to prevent the code and any output from. Code chunk would look like `{r chunk_name, include=FALSE}`
   - Utilize `warning=FALSE` and `message=FALSE` to suppress any warnings/messages that may be displaying from any chunks
   - For this milestone please make sure code for visualizations are included in the final file. Show your work by "echoing" code used in Rmd/Qmd file to create your tables and graphs. "'{r, echo = TRUE} in code chunk preferences.
   - Use headers and sub headers to create an organized document
   - Make sure you commit and push your rsconnect folder! Read more details here.Links to an external site.